Lean proof simplification after exact order and scalar constructions

A fifth review with all earlier proposals assumed available

Source commit `9f35798db4a63c25e435fca14d75931144657d7e`

1 Findings and recommendation

The useful fifth iteration is a small proof-only cleanup, not another general framework. After excluding every earlier proposal, the new exact-order and padding clients still contain avoidable proof transport. The core packet removes 24 physical lines across three complete source files, adds no helper or import, and changes no public statement or definition. The recommended next action is to compile these small patches independently, not to enlarge the abstraction layer.

Candidate	Full-file lines	Treatment
Guarded power-cast equivalence	$80 \to 73$; -7	Proof-only trial
Exact-order clients and modulus transport	$105 \to 91$; -14	Proof-only trial
Padding residue-to-bit client	$224 \to 221$; -3	Narrow local cleanup
Core total	$409 \to 385$; -24	No new helper cost
Positivity from coprimality	$201 \to 198$; -3	Optional dependency tradeoff

The optional three-line saving is not an unconditional recommendation. Deriving `Gamma_pos` from a calibrated coprimality theorem makes positivity depend on the literal-kernel residue chain. Its current elementary proof avoids that dependency. The same source-length caution applies to replacing a direct residue endpoint by a stronger exact-order result: shorter text is not automatically better dependency structure or faster compilation.

Current source progress is not a new proposal here. The literal K residues and all three corresponding H_s residues are now proved. Exact auxiliary prime-power order, full affine power coordinates, actual rank-step units and constructed mask/scalar consequences are also in the inspected source. In particular, this report does not carry forward the fourth review's open literal-K status. Those developments create the fresh clients being reviewed; their construction is not counted as this packet's saving.

Every new Lean candidate remains uncompiled. No Lean, Lake or Elan executable is available in this task environment, and no toolchain was installed. Static source checks, exact patches, independent mathematical explanations and finite tests are included; none establishes Lean elaboration, kernel acceptance, an axiom footprint or a performance improvement. The PDF is compiled separately.

The pinned main tree contains 686 Lean files and 85,917 physical lines. Novelty is measured against 36 placed manuscripts at the final bounded exclusion cutoff, including the relocated incoming archives, fourth packet and seven fresh manuscripts. Earlier contracts are assumed available even when optional, deferred or mutually incompatible. The review is deliberately modest: it does not claim a new general theorem or completion of the full polynomial–Con(ZFC) equivalence.

2 New source and excluded prior work

This review starts after the fourth packet, not before it. Its pinned tree is `9f35798db4a6`; the prior exclusion cutoff is `663fb267f614`. At the source pin, the two proposal folders contain 28 direct manuscripts: 23 placed packet trees, a–w, and five incoming archives. Their tree identities and archive bytes are unchanged. Only the adoption ledger changed inside those folders. The separately delivered fourth packet is also excluded, but is not in that initial 28-manuscript census.

Final bounded exclusion check. At `91dce223cda8`, there are 36 placed manuscripts and no incoming archives. Thirteen new placements comprise the five old incoming archives, the fourth packet now placed as zh, and seven fresh packets: zc, zd, ze, zf, zg, zi and zj. All are assumed available. All four target files and 15 dependency files are byte-identical at this later cutoff; the candidate source pin remains fixed.

Meaning of assumed applied. Every earlier useful contract is available for novelty analysis, including deferred, optional, uncompiled and mutually incompatible alternatives. This is a counterfactual review baseline, not a claim that all old implementations coexist in a compiled tree. Moving an old argument into a new helper, new record or new filename earns no new credit. A fresh client must still eliminate residual work after selecting the relevant old contract.

Prior exclusions and provenance are retained under `catalog/`; the earlier report is treated as baseline, not repackaged. It already delivered finite-chain order-isomorphism rigidity, the `Fin.pred` predecessor option, the exact coprime-cardinality client, inlining of the three conditional H_s residue clients, and the exact marker-only terminal band. Packet p already contains the castHom/value projection proposed and withdrawn late in round four. None is re-proposed here.

What has genuinely changed in the source. Four developments materially alter what can now be reviewed:

1. The original literal K now has proved residues 26, 1995 and 114787 modulo 47, 2011 and 120199. Consequently the three actual H_s residues are unconditional: 29, 674 and 36379. The older conditional wrappers survive, but the previous open literal-K obligation is closed at this pin. See `CurrentKernelResidues2011.lean`:798–841.

2. Exact principal-unit order is proved through LTE and a nontrivial checked lift. The actual full controller modulus then supplies affine power coordinates, rather than assuming an orbit coordinate map. See `PrimePowerOrder.lean`:40–78 and `CurrentPowerCoordinates2011.lean`:54–104.

3. The actual rank-step coefficient is proved a unit for independent auxiliary-power width and cell dimensions, including zero exponents. See `CurrentControllerUnits2011.lean`:22–27 and `CurrentControllerUnits2011.lean`:181–199.

4. New padding, origin, high-mask and scalar-witness modules derive consequences of the actual normalized constructed word. Same-source and same-carry binding is essential; arbitrary numeric maps may not silently become bits. See `CurrentWordHighMask2011.lean`:75–107 and `CurrentWordScalarWitness2011.lean`:109–154.

Upstream adoption is not this packet's saving. The ledger records five more adopted families with a combined net reduction of 228 lines, and 3016 lines across its reviewed adoption series. These are repository-reported totals, not independent reruns or new fifth-review credit. The bounded source interval has 14 intervening commits, 60 changed repository files, and 42 changed active Lean files with $+2679/-372$ physical lines. Merge commits are not independent changes.

3 Fuse the guarded power cast equivalence

Remove duplicate implications, not the natural-subtraction guard. In PrimePowerOrder.lean:26–37, `pow_cast_eq_one_iff` proves

$$(g : \mathrm{ZMod}(m))^k = 1 \quad \Longleftrightarrow \quad m \mid (g^k - 1), \qquad 1 \le g.$$

The source proves both directions by converting the natural difference to a zero cast. The proposed proof rewrites the equivalence itself once:

```
theorem pow_cast_eq_one_iff {g m : ℕ}
    (hg : 1 ≤ g) (k : ℕ) :
    (g : ZMod m)^k = 1 ↔ m ∣ g^k-1 := by
  rw [← CharP.cast_eq_zero_iff (ZMod m) m (g^k-1),
    Nat.cast_sub (one_le_pow₀ hg), Nat.cast_pow,
    Nat.cast_one, sub_eq_zero]
```

The exact patch keeps the existing two-line statement and uses two proof-body lines. The declaration changes from 11 to 4 physical lines; the full module changes from 80 to 73. There is no new helper, wrapper, import or definition edit. The use of a standard rewriting chain is routine; the saving is elimination of the two separate copies in this new exact client.

Mathematical justification. From $1 \le g$, obtain $1 \le g^k$. Thus the natural subtraction is untruncated and its cast equals $(g : \mathrm{ZMod}(m))^k - 1$. In characteristic m, a natural number casts to zero exactly when m divides it. Finally $u - 1 = 0$ is equivalent to $u = 1$. These are the same facts already used in the original forward and reverse proofs.

Why the guard cannot disappear. Without $1 \le g$, take $g = 0$, $m = 2$ and $k = 1$. In natural arithmetic $g^k - 1 = 0$, so the divisibility statement is true, but $0 = 1$ in ZMod(2) is false. The replacement retains `hg` exactly, rather than allowing a cast tactic to conceal this truncation issue.

Boundary cases remain included. The theorem retains arbitrary m and k, including $m = 0$, $m = 1$ and $k = 0$. No positivity or nonzero instance is added for the modulus. The $m = 0$ case is interpreted through ZMod(0), not by dividing by zero in an integer remainder test. The finite checker uses that separate interpretation.

The downstream LTE theorem still requires an odd prime p, $p \mid (g - 1)$ and $p^2 \nmid (g - 1)$; it establishes an exact valuation, not merely a lower divisibility bound. The exact order theorem still handles the modulus p through its $k = 0$ branch, and `order_prime_power` still requires $1 \le n$. None of these hypotheses is weakened or re-proved here. See PrimePowerOrder.lean:40–78.

Validation to perform in Lean. Elaborate the rewritten equivalence first, then the existing prime-power checks and actual-generator clients. In particular, verify the orientation of `CharP.cast_eq_zero_iff`, the inferred natural power bound, and simplification of the final reflexive equivalence. The mathematical argument is complete; elaboration is not established by that argument or by the supplied finite tests.

4 Reuse the actual exact order results

The new full-modulus file contains small proofs that can use its own stronger results. The bundle changes `CurrentPowerCoordinates2011.lean` from 105 to 91 physical lines. It combines three local cleanups, with no new imports or helpers and no public signature changes.

CRT order calculation. The existing Chinese remainder ring equivalence preserves multiplicative order, and a product's order is the least common multiple of the two component orders. Inline the intermediate image equality directly at its only use:

```
  calc
    _ = orderOf (ZMod.chineseRemainder hab
        ((x : ZMod (a*b))^k)) :=
      ((ZMod.chineseRemainder hab).toMulEquiv.orderOf_eq _).symm
    _ = _ := by
      rw [map_pow, map_natCast]
      exact Prod.orderOf_mk
```

The exact declaration changes from 12 to 7 physical lines. The coprimality premise is unchanged, with no extra nonzero-modulus assumptions. The pinned APIs are `Mathlib/GroupTheory/OrderOfElement.lean:368–377` and `Mathlib/GroupTheory/OrderOfElement.lean:1388–1393`.

Two order-one endpoints. Specializing the already proved actual order formulas at exponent zero yields order one, hence the generator is one:

```
-- generator_cast_auxiliary
  apply orderOf_eq_one_iff.mp
  simpa only [zero_add, pow_one, pow_zero] using
    generator_order_succ 0
-- generator_cast_C0
  apply orderOf_eq_one_iff.mp
  simpa only [zero_add, pow_one, pow_zero] using full_order 0
```

The first declaration saves one line; the second saves two. The decisive equivalence is `Mathlib/GroupTheory/OrderOfElement.lean:254–260`. There is no circularity: `full_order` uses the fixed-factor identity and `generator_order_succ`; it does not depend on either endpoint being replaced. The exact upstream statements are in `CurrentAuxiliaryOrder2011.lean:52–60` and `CurrentPowerCoordinates2011.lean:46–64`.

These shorter endpoint proofs intentionally depend on the stronger exact-order development, whereas the old auxiliary endpoint had a direct residue proof. No import is added, and the actual calibration is fixed, but maintainers may reasonably keep the lower-level proof if independence from LTE is important.

Transport only the small modulus expression. Replace the one-use private `order_transport` by

```
  simpa only [modulus_factorization] using full_order k
```

inside `block_order`. Its three-line declaration remains three lines. Deleting the four-line helper, its one-line doc comment and one blank line saves six physical lines; no helper cost is hidden. Together the bundle saves $5 + 1 + 2 + 6 = 14$ lines.

Preserved scope. Every modulus, public wrapper, opacity guard and actual coordinate conclusion stays unchanged; the transport never evaluates the huge base. Generic finite-surjectivity, coordinate records and the old castHom/value projection are excluded. Late packet zd's finite prime-factor exact-order certificates are also baseline. They do not remove the CRT, zero-level or equality-transport steps targeted above.

5 An optional positivity proof with a dependency cost

Three lines can be removed, but the old proof has a useful independence property. The exact client is `CurrentControllerUnits2011.lean:195–199`. It proves $0 < \Gamma(r_w, n)$ for every pair of natural exponents, including $r_w = 0$ and $n = 0$. The old proof multiplies three positive factors. The replacement observes that the already proved coprimality with 47 excludes zero.

```
theorem Gamma_pos (rw n : ℕ) : 0 < Gamma rw n :=
  Nat.pos_of_ne_zero fun hz => by
    simpa [hz] using coprime_Gamma_47 rw n
```

The supplied patch keeps the inner proof on one line, so the declaration occupies two physical lines instead of five. The listing is wrapped for reading. The full module changes from 201 to 198 physical lines; it adds no helper, import or wrapper. Every definition and theorem statement is unchanged.

Mathematical justification. For natural x, $\gcd(0, 47) = 47 \neq 1$. Thus $\mathrm{Coprime}(x, 47)$ implies $x \neq 0$, hence $0 < x$. This uses the exact existing theorem `coprime_Gamma_47`; it neither generalizes the unit theorem nor proves new calibration data. The original factors are

$$\Gamma(r_w, n) = \rho^{977}\big(q(n)^2 - 1\big)\big(1 + q(n)A(r_w)\big),$$

where $\rho = 2^{47}$, $q(n) = B^{p^n}$, $A(r_w) = K + B^{p^{rw}}$, and $p = 120199$. The original proof obtains positivity from $q(n) > 1$, $\rho > 0$ and the final factor being at least one; it does not need any residue of K.

The tradeoff is real. The replacement depends on `coprime_Gamma_47`, whose proof goes through the nonzero quadratic and coupling residues. The latter uses the calibrated residue of the original literal kernel. Positivity therefore acquires a dependency on the full current calibration chain even though its elementary mathematical reason does not require that chain. This is not an additional assumption in the theorem statement, but it matters for modular maintenance and which intermediate files can be reused independently.

Recommendation. Treat this as an optional consequence-reuse variant, not a default improvement. Keep the elementary proof if positivity is intended to remain independent of kernel calibration. Adopt the shorter proof only if that dependency is acceptable and the compiled proof is at least as clear to maintainers. No runtime benefit is asserted.

Preserved distinctions. The two dimensions remain independent. Neither exponent is strengthened to be positive. The natural subtraction $q(n)^2 - 1$ still agrees with the signed rank expression only through its retained lower-bound proof; the replacement does not remove `quadratic_ge_one`. The definitions of K, B, q, A and Γ and their local opacity attributes are untouched. In particular, a proof of a unit does not itself construct an orbit or certify a safe controller mutation.

The unit proof chain and full-width modulus remain as inspected in `CurrentControllerUnits2011.lean:146–199`. This packet does not count prime-coprimality helper reuse, Frobenius, scalar rank factorization or cast-to-residue bridges as new abstractions.

6 A narrow bit test cleanup without changing the word

Keep the current mask construction and shorten only its residue-to-bit client. The retained target is `CurrentWordPadding2011.lean:125–133`. For an arbitrary natural x,

$$x \bmod 2048 \le 5 \quad \Longrightarrow \quad x \,\&\, 8 = 0.$$

The source first clears bit three of the residue, separately identifies that bit with the bit of x, and transfers the result. A restricted simplification performs the same transfer directly.

```
theorem bit3_clear_of_residue {x : ℕ}
    (hx : x%2048 ≤ 5) : x &&& 8 = 0 := by
  simpa only [PaddingStaircase.bit3_mask_iff,
    Nat.testBit_mod_two_pow,
    show (3 < 11) = True from propext (by decide),
    decide_true, Bool.true_and] using
    (Nat.testBit_lt_two_pow (show x%2^11 < 2^3 by omega))
```

The patch uses three proof-body physical lines instead of six; the listing is wrapped for legibility. The complete module changes from 224 to 221 physical lines, with zero helper or import cost. All bytes outside this proof body are retained.

Why it is correct. Let $r = x \bmod 2^{11}$. The premise gives $r \le 5 < 8 = 2^3$, so bit three of r is false. Reduction modulo 2^{11} preserves every bit below eleven, in particular bit three. The established mask equivalence then identifies $x \,\&\, 2^3 = 0$ with that false bit. The proof does not infer $x < 8$: values $x = 2048t + r$, with arbitrary t and $0 \le r \le 5$, remain in scope.

Novelty boundary. This is a new-client deletion, not a new bit-test method. The source already uses the same `testBit_mod_two_pow` simplification pattern in `PaddingStaircase.lean:23–33`. That existing technique is treated as available. What remains here is the removable intermediate transport in a newly added client, not a new mask theorem, carry architecture or scalar witness interface.

The surrounding semantic guards stay intact. This patch leaves the padding support bounds, reflected residue calculation, zero marker input and same-source carry normalization untouched. It does not turn arbitrary numeric maps into Boolean maps, discard matching or origin hypotheses, or replace an actual normalized word by an unrelated witness. The assembled padding conclusion continues to refer to the same word, carry and digits supplied in `CurrentWordPadding2011.lean:179–222`.

Rejected scalar shortening. The candidate that rewrote $q = \rho^{N \cdot 2011 - 1}\rho$ directly with predecessor-power cancellation is excluded: earlier proposal E41 already supplies exactly that mechanism in packet f. It is not included in the reported saving. Generic quotient/remainder packages, signed-prefix borrowing, coherent source/carry views, source-residue zeros and the fourth packet's marker-only terminal band are also baseline material, not fresh recommendations.

The scalar witness definitions and the positive transport factor remain unchanged. In particular, the actual signed transport factor is one, not zero; its natural quotient witness is a successor, and all cast-subtraction guards must remain available.

7 Count complete files and keep optional costs separate

The core saving is 24 physical source lines, not a theorem or performance count. All three core patches are against the same source pin and edit disjoint files. The complete before and after files are supplied, so the totals include imports, comments, blank lines, namespace scaffolding and retained compatibility declarations.

Complete file	Before	After	Removed
`PrimePowerOrder.lean`	80	73	7
`CurrentPowerCoordinates2011.lean`	105	91	14
`CurrentWordPadding2011.lean`	224	221	3
Core	**409**	**385**	**24**
`CurrentControllerUnits2011.lean`	201	198	3
Core plus optional positivity	610	583	27

Helper accounting. No new Lean helper, import, wrapper or public declaration is added. The exact-order bundle deletes the one-use private `order_transport`: four declaration lines, one doc-comment line and one blank line. Its client is rewritten in place and stays three lines. The combined 14-line saving is not a four-line body estimate: it includes the complete helper deletion and every changed client.

The guarded cast declaration contributes seven lines. In the exact-order file, direct CRT composition contributes five, the auxiliary order-one specialization contributes one, the full fixed-factor specialization contributes two, and helper removal contributes six. The padding proof body contributes three. The optional positivity declaration contributes three, partly through terse layout; its logical dependency change is more important than that cosmetic count.

Secondary measures are not interchangeable. The independent audit records 23 nonblank lines removed in the core and three more in the optional variant. The power bundle alone removes 20 nonblank lines as well as 21 physical lines. The mask checker records 175 to 172 comment-stripped nonempty code lines and 2587 to 2553 regex lexical tokens for the full module. Those tokens are not Lean parser tokens. No aggregate token count is inferred across differently defined metrics. The physical-line total above is the common reproducible measure.

No double counting. Existing helpers and the already imported exact-order infrastructure are baseline costs, not new additions. The old 228-line adoption increment and 3016-line ledger total are not included. The withdrawn scalar prefix candidate receives zero saving. The optional positivity variant is separate, and all-source totals are not presented as default adoption advice.

Evidence cost is visible but not proof-source cost. The report, before/after snapshots, patches, manifests, independent verification scripts and captured results are review artifacts, not Lean library additions. They are delivered separately rather than silently counted as zero-size files. No new Lean regression module is claimed: the sequential validation plan uses the existing checks and downstream targets.

Definition boundary. These are proof-only edits. No public definition body changes, no stronger preconditions, and no theorem renaming are proposed. In particular, the literal huge constants remain opaque where the source already protects them, and the exact original kernel is not replaced by a congruent stand-in.

8 Validate in small sequential steps

Static validation passed; Lean validation is still the acceptance gate. The candidate verifiers check pinned hashes, exact patches, signatures and non-target bytes. The table records their finite runs, not universal proofs or elaboration; section 9 covers the independent audit.

Check family	Recorded finite scope
Guarded cast equivalence	27,885 cases; zero modulus/exponent and missing-guard counterexample
CRT order and prime powers	27,265 CRT cases (266 zero-modulus); 144 odd-prime order cases
Modulus factorization	512 cases; actual-generator residues computed without expanding the huge base
Padding bit transfer	20,480 cases; 60 guarded; high multiples and guard mutations
Optional unit consequence	147 modular cases from supplied calibration, not independent literal-K evaluation

Build against the declared versions. Use `lake build` with fully qualified targets, Lean `v4.32.0` and Mathlib revision `81a5d257c8e410db227a6665ed08f64fea08e997`. Start from a clean pinned worktree with its existing dependencies available. Record a successful unmodified baseline before applying a patch. Installation or cache retrieval, if needed, is a separate environment action; this review did neither.

Build process safety. The repository warning says `LAKE_JOBS=1` is ineffective here. Run one Lean process/module at a time, only after its dependencies are compiled; rebuild each main module before its Checks module. Never launch an umbrella build over stale dependencies, and never set `LEAN_NUM_THREADS=0`.

1. Apply only `guarded_power_cast_fusion.patch`. Rebuild the `PrimePowerOrder` and `CurrentAuxiliaryOrder2011` mains in dependency order, then their Checks modules. Inspect the exact cast proof, odd-prime/LTE guards and $k = 0$ endpoint before proceeding.

2. Apply `exact_order_clients.patch`. Rebuild `CurrentPowerCoordinates2011` first, then its Checks module and the generic `ControllerPowerCoordinatesChecks`. Confirm the shortened CRT map computes definitionally to the product, both zero-exponent specializations elaborate, and dependent modulus transport keeps the base opaque.

3. Apply `bit3_clear_of_residue.patch`. Build the padding checks, then the origin, high-mask and scalar-witness checks. Confirm the original statement is unchanged and the restricted simplifier transfers the low bit without adding a bound on the full digit.

4. Only if the dependency tradeoff is accepted, apply `gamma_pos_from_coprime.patch` and build `CurrentControllerUnits2011Checks`. Otherwise keep the elementary positivity proof.

5. Only after all dependencies are compiled, build the encompassing `DiophantineConsistency` target, inspect the exact diff, and rerun the original downstream checks. If adopting on the later exclusion-cutoff tree, also build `CurrentControllerSubsets2011Checks`: its new subset client uses `block_order`. Record versions, commands, exits and failures. Compare `#print axioms` output before and after if the project's acceptance policy requires it; no axiom equality is asserted here.

9 Evidence and limits

Reproducibility begins with identities. The source pin is `9f35798db4a63c25e435fca14d75931144657d7e`. The repository declares Lean `v4.32.0`; its manifest pins Mathlib at `81a5d257c8e410db227a6665ed08f64fea08e997`. Candidate source links resolve to the source pin, and library links to that Mathlib revision, not to a moving branch. The later exclusion cutoff is `91dce223cda86c0755b8a9b2504f84be04443fce`. The local evidence records file hashes and Git blobs as well as line anchors.

Evidence	What it establishes
`catalog/corpus_inventory.json`	Exact identity of the 23 placed trees and five archives through the source pin
`catalog/late91dce/`	Final placement and fresh-manuscript comparison at the later cutoff
`catalog/inherited_evidence_manifest.json`	Hashes of the inherited semantic catalog and separately delivered fourth report
`catalog/EXCLUSION_AND_ADOPTION_LEDGER.md`	Assumed-contract baseline, new adoption status and candidate-screen boundaries
`catalog/fresh_source_inventory.json`	Fresh module hashes, Git blobs, line counts and declaration anchors
`catalog/source_evidence.json`	Pinned source excerpts supporting changed mathematical scope
Candidate notes, full after-files and diffs	Exact proposed source delta and helper cost; not elaboration evidence
Audit and validation records	Independent static checks and explicitly bounded non-Lean tests

Independent integrated audit. Source and finite sanity checks passed with the core patches in all six orders and all four patches in all 24 orders. Normal and optimized Python runs agreed; the isolated replay made no file writes. Six coherent candidate-plus-patch mutations were rejected in both modes. This tests the checker and integration boundary, not Lean validity. See `audit/integrated_results.json` and `audit/README.md`.

Novelty is semantic. The targeted name searches are supplementary: absence of a spelling is not evidence that an idea is new. The full inherited exclusions include fixed-state scanner recursors, cyclic residue and quotient normalization, signed prefix borrow, coherent word/carry views, mask-block lookup, source/witness extraction, finite invariant predecessors and generic rank interfaces. Those mechanisms remain free baseline contracts for judging the small current-client changes.

What was not established. No new Lean candidate was elaborated or checked by the kernel here. No additional axiom claim is certified, no compile-time benchmark was run, and no memory or runtime improvement is inferred from a shorter proof. Existing repository compilation and axiom-review statements were read, not rerun. PDF compilation and finite arithmetic checks are separate evidence classes.

Endpoint boundary at the comparison cutoff. The newer subset construction still separates physical placement and the rank factor, while ordinary dummy transport retains a nonwrapping bound. These are explicit limits of those theorems, not a claim about commits after the cutoff. The scalar module remains partial. This report establishes neither the full polynomial–Con(ZFC) equivalence nor a change to its trust boundary.

Recommendation. Only adopt candidates that compile against the pinned dependencies and preserve their public contracts. If a proof fails, repair that proof or drop its patch; do not change the theorem to fit the tactic.